

Analysis of Swarm Intelligence Metaheuristics for Mitigating Routing Attacks in VANETs

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Abstract—Vehicular Ad-Hoc Networks (VANETs) are characterized by high mobility, dynamic topologies, and strict Quality of Service (QoS) requirements, making them highly susceptible to routing attacks. This report aims to perform an analysis of three major Swarm Intelligence (SI) techniques: Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), and the Firefly Algorithm (FA) and their impact on routing security and network reliability. By evaluating related academic papers, this study compares their efficacy in attack detection and their impact on QoS and network performance metrics, in particular the Packet Delivery Ratio (PDR) and End-to-End Delay. The findings suggest that while ACO excels in path redundancy for smaller networks, PSO is superior for low-latency parameter optimization, and Firefly is the most suitable in dense urban environments.

I. INTRODUCTION

A. Background

VANETs are a subgroup of Mobile Ad-Hoc Networks (MANETs) designed to enable Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication. The primary goal is to enhance road safety and traffic efficiency. However, the lack of a centralized infrastructure and the "open" nature of wireless communication invite malicious actors to disrupt the routing process. Traditional methods often struggle with the low-latency requirements of vehicular safety applications, leading researchers toward bio-inspired Swarm Intelligence (SI) to provide adaptive, decentralized security. SI is a branch of Artificial Intelligence (AI) inspired by the collective, decentralized behavior of social animals such as ants, bees, birds, and fish. In these systems, individual members (often called "agents") follow very simple, local rules. Even though there is no central "leader" or master plan, the group as a whole exhibits highly sophisticated, "intelligent" behavior that allows them to solve complex problems, such as finding the shortest path to food, escaping predators, or building intricate nests.

B. Scope of Review

This report focuses on the application of SI metaheuristics to the routing layer of VANETs. It specifically analyzes how these algorithms identify malicious nodes and reroute traffic to maintain QoS. The scope is limited to a comparative

analysis of three specific peer-reviewed papers representing ACO, PSO, and Firefly methodologies.

II. PROBLEM DEFINITION

A. Problem Statement

The dynamic nature of VANETs creates a moving target for security protocols. Malicious nodes can easily join the network and execute routing attacks that drop, delay, or misdirect critical safety data. Existing standard routing protocols (like AODV or DSR) are "trusting" by nature and cannot inherently distinguish between a link failure due to high mobility and a deliberate attack.

B. Motivation

Each swarm algorithm has its advantages and disadvantages and can be applied to different networking scenarios. Motivation for this study stems from the need to categorize which SI technique is best suited for specific use cases.

III. BACKGROUND

A. Routing Attack Types in VANET

To evaluate the effectiveness of SI, one must define the threats they are designed to mitigate:

- **Black Hole Attack:** A malicious node advertises the shortest path to a destination but drops all intercepted packets instead of forwarding them, leading to a massive drop in PDR.
- **Wormhole Attack:** Two colluding nodes capture packets at one point and tunnel them to another location, confusing the network topology.
- **Grey Hole Attack:** A variation of the Black Hole where the node selectively drops packets (e.g., only UDP packets or specific IDs), making detection difficult.
- **Sinkhole Attack:** An intruder tries to attract all traffic from a specific area through itself by spoofing routing information.

- Sybil Attack: A single node assumes multiple identities (pseudonyms) to simulate a traffic jam or outvote honest nodes in trust-based systems.
- Illusion Attack: The attacker broadcasts false sensor data (e.g., a fake accident) to manipulate surrounding vehicles.
- Denial of Service (DoS): Flooding the network with useless traffic to exhaust bandwidth and prevent safety messages from being delivered.

IV. SELECTED ARTICLES (NOTE PAPERS SUBJECT TO CHANGE IN FINAL REPORT)

A. Ant Colony Optimization (ACO) Reference: *Ant Colony Optimization Based Approach for the Detection of Black Hole Attack in WANET (IJCSMC, Vol. 4).*

Ant Colony Optimization (ACO) is inspired by the foraging behavior of ants. When ants find food, they leave a chemical trail called pheromones. Other ants detect these trails and follow them, reinforcing the pheromone levels. In VANETs, ants are control packets searching for paths and pheromones represent path quality (low delay, high trust). If a path starts dropping packets (Black Hole), the "pheromones" evaporate, and the swarm naturally shifts to safer, more reliable routes.

- Simulation and Methodology: The researchers utilized MATLAB R2010a to simulate a network of 50 mobile nodes. The algorithm initializes with a population of 500 ants over 10 iterations [1]. It introduces a pheromone-based trust system where paths are reinforced based on a combination of Bit Error Rate (BER) and successful delivery.
- Key Results: Under a Black Hole attack, the standard Dynamic Source Routing (DSR) protocol failed significantly. Applying ACO resulted in a 15% improvement in PDR for DSR [1]. More notably, when applied to the Optimized Link State Routing (OLSR) protocol, ACO-based optimization resulted in a performance increase of 54% in terms of overall network efficiency and error reduction. The research demonstrated that under attack, the standard PDR dropped to 12%. The ACO-optimized network achieved a 95% PDR [1].
- Conclusion: ACO was most effective at providing redundant paths that automatically bypass nodes with high Bit Error Rates (BER) [1].
- Additional Detail: ACO proved highly effective at "cleansing" the route selection process by lowering the BER on paths containing suspicious nodes.

B. Particle Swarm Optimization (PSO) Reference: *Particle Swarm Optimization-Based Content Delivery Protocol for UAV-VANETs (IEEE/MDPI).*

Particle Swarm Optimization (PSO) is modeled after the social behavior of bird flocking or fish schooling. Each

potential solution is a particle moving through a search space. Important parameters to note include Personal Best (P_{best}): The best solution an individual vehicle has found and Global Best (G_{best}): The best solution found by any vehicle in the entire swarm. Particles adjust their "velocity" and "position" (security parameters) toward the most successful nodes in the network. It is extremely fast at finding the "optimal" threshold for detecting anomalies.

- Simulation and Methodology: This research integrates PSO to optimize the selection of Cluster Heads (CH) in a hybrid UAV-VANET environment. The fitness function for each particle (vehicle) is calculated based on Residual Energy, Node Mobility (Velocity), and Distance to Neighbors.
- Key Results: PSO demonstrated superior computational speed. The protocol reduced End-to-End Delay by 25% compared to traditional AODV [2]. In high-mobility scenarios (up to 120 km/h), the PSO-optimized swarm maintained a stable PDR of 88%, ensuring that content delivery was not interrupted by rapid topology changes [2].
- Conclusion: PSO is a suitable method for time-sensitive safety applications where reducing latency is the primary goal.
- Additional Detail: PSO's strength lies in its ability to converge on an optimal trust threshold globally, rather than just locally at the node level.

C. Firefly Algorithm (FA) Reference: *BSHR-FA: A Blockchain-Enabled Secure Hierarchical Routing Using Firefly Algorithm for VANETs (Security and Privacy, Wiley, 2025).*

The Firefly Algorithm mimics the flashing behavior of fireflies to attract mates or prey. In VANETs, brightness represents a node's trust level or signal strength. Dimmer fireflies move toward brighter ones. Malicious nodes (Black Hole/Sybil) are assigned a low brightness because of their poor performance. Legitimate nodes are naturally attracted to bright, high-performing nodes, creating secure clusters that isolate intruders.

- Simulation and Methodology: This study uses FA for secure hierarchical routing. Brightness (I) is defined by a node's trust score and available bandwidth. Fireflies (nodes) are attracted to others with higher brightness, effectively forming clusters around the most trustworthy nodes [3].
- Key Results: In dense simulations of 150+ nodes, this protocol maintained a 90% PDR [3]. Furthermore, the integration with blockchain allowed for a 98% detection rate of Sybil and Wormhole nodes, as malicious nodes could not replicate the brightness required to be selected as a routing hub [3].
- Conclusion: Firefly is an appropriate choice for large-scale urban grids where scalability and high detection accuracy are required.

- Additional Detail: FA's random-walk component helped it avoid local optima traps, making it more resilient to sophisticated Grey Hole attacks.

V. DISCUSSION

A. Strengths and Limitations

- ACO is the most reliable for finding alternate physical routes in small-to-medium networks, but pheromone evaporation can lag during extreme high-speed transitions.
- PSO provides the fastest convergence for parameter tuning, making it the best for reducing latency. However, it can occasionally be deceived by Grey Hole attacks if the global best position is influenced by biased data.
- Firefly offers the most stable structure for large, dense networks. Its hierarchical approach prevents networks from being overwhelmed by excessive control packets which is a problem common in ACO, though its mathematical complexity is higher.

VI. CONCLUSION

A. Summary

This report evaluated ACO, PSO, and Firefly algorithms against routing-layer threats in VANETs. Through a meta-analysis of three distinct papers, it was demonstrated that while all three significantly improve PDR compared to unprotected networks, they serve different operational purposes. ACO is the most resilient for path discovery, PSO is the most efficient for latency, and Firefly is the most scalable and accurate for dense environments.

B. Future Work

Future research should focus on Hybrid Swarm Intelligence (HSI). Combining ACO's pathfinding with PSO's parameter tuning could create a protocol that is both fast and resilient. Additionally, the integration of these swarm techniques with Machine Learning (ML) could allow the network to predict an attack before it occurs.

REFERENCES

- [1] Ant Colony Optimization Based Approach for the Detection of Black Hole Attack in WANET, International Journal of Computer Science and Mobile Computing (IJCSMC), Vol. 4.
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